LEAP: Supplementary Proofs

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***Abstract*—This repository companion collects the complete proofs of LEAP's cross-field lower bound, critical-position proximity property, weighted DTW block bound, and DTW signal lower bound. The four appendices correspond to Appendices A–D of the repository README. All statements concern real arithmetic; the implementation links identify the corresponding computations.**

NOTATION

For a trajectory T, let V_T be its sampled positions and E_T the union of the line segments joining consecutive positions. The distance from a point x to a set S is $\operatorname{dist}(x, S) = \min_{z\in S}\|x - z\|_2$. For probe p_j^{pr}, the nearest-position and segment fields are

$$\begin{aligned} a_j(T) &= \operatorname{dist}(p_j^{\mathrm{pr}}, V_T), \\ b_j(T) &= \operatorname{dist}(p_j^{\mathrm{pr}}, E_T). \end{aligned} \tag{1}$$

The directed and symmetric vertex-to-segment Hausdorff distances are

$$\begin{aligned} h(T_A, T_B) &= \max_{p\in V_{T_A}} \operatorname{dist}(p, E_{T_B}), \\ D_H(T_A, T_B) &= \max\{h(T_A, T_B), h(T_B, T_A)\}. \end{aligned} \tag{2}$$

The cross-field lower bound combines all probes and both directions:

$$\begin{aligned} L_H(T_A, T_B) = \max\big\{0,\ &\max_j[b_j(T_B) - a_j(T_A)], \\ &\max_j[b_j(T_A) - a_j(T_B)]\big\}. \end{aligned} \tag{3}$$

APPENDIX A
VALIDITY OF THE CROSS-FIELD BOUND

Proposition 1: For any probe positions, $0 \le L_H(T_A, T_B) \le D_H(T_A, T_B)$ in real arithmetic. Validity does not depend on learning.

Proof: Fix p_j^{pr} and choose its nearest position $p \in V_{T_A}$. Apply the Euclidean triangle inequality through p to its nearest point on E_{T_B} to obtain

$$\begin{aligned} b_j(T_B) &= \operatorname{dist}(p_j^{\mathrm{pr}}, E_{T_B}) \\ &\le \|p_j^{\mathrm{pr}} - p\|_2 + \operatorname{dist}(p, E_{T_B}) \\ &\le a_j(T_A) + h(T_A, T_B). \end{aligned} \tag{4}$$

Hence $b_j(T_B) - a_j(T_A) \le h(T_A, T_B)$. Interchanging T_A and T_B proves the other direction. Taking the maximum over probes and both directions, with zero included, yields the result. ∎

APPENDIX B
CRITICAL-POSITION PROXIMITY

A *critical position* p^* is a sampled position that attains $D_H(T_A, T_B)$ in either direction. For probe set Θ, let $\delta_\Theta(p^*) = \min_j \|p_j^{\mathrm{pr}} - p^*\|_2$ be its distance to the nearest probe.

Proposition 2: For any critical position p^*,

$$0 \le D_H(T_A, T_B) - L_H(T_A, T_B) \le 2\delta_\Theta(p^*). \tag{5}$$

Proof: Suppose $p^* \in V_{T_A}$ and $\operatorname{dist}(p^*, E_{T_B}) = D_H(T_A, T_B)$. Choose the nearest probe p_j^{pr}, at distance $\delta = \delta_\Theta(p^*)$. By the same triangle-inequality argument, $b_j(T_B) \ge D_H(T_A, T_B) - \delta$. Since p^* is a position of T_A, $a_j(T_A) \le \delta$. Therefore

$$\begin{aligned} L_H(T_A, T_B) &\ge b_j(T_B) - a_j(T_A) \\ &\ge D_H(T_A, T_B) - 2\delta. \end{aligned} \tag{6}$$

The case $p^* \in V_{T_B}$ follows by symmetry, and validity gives nonnegativity. ∎

APPENDIX C
WEIGHTED DTW BLOCK BOUND

This bound is a check before original-distance refinement, not the signal bound used for candidate ranking. It follows the paper's DTW subsection under Filtering and Exact Search.

A. Blocks and Costs

Let $q = (p_1^q, \ldots, p_n^q)$ and $T = (p_1^T, \ldots, p_m^T)$ be nonempty trajectories. Their DTW distance $D_W(q, T)$ is the minimum sum of Euclidean point-to-point distances along an alignment from $(1, 1)$ to (n, m), using steps $(1, 0)$, $(0, 1)$, or $(1, 1)$. Thus every position of each trajectory is matched at least once, and positions are visited in order.

We split each trajectory into four consecutive blocks with nearly equal position counts. For a trajectory of length $n \ge 4$, block s contains positions $\lceil (s-1)n/4 \rceil + 1$ through $\lceil sn/4 \rceil$, for $s \in \{1, 2, 3, 4\}$. If $n < 4$, each position forms one block and the remaining slots are empty. Query block s contains n_s positions, and candidate block t contains m_t positions.

For each nonempty block and fixed filtering probe $p_{W,j}^{\mathrm{pr}}$, we record the minimum and maximum distance from its positions to that probe. The intervals are $[l_{sj}^q, h_{sj}^q]$ for the query and $[l_{tj}^T, h_{tj}^T]$ for the candidate. Rows of the following matrix index query blocks; columns index candidate blocks:

$$G_{st} = \max_j \max\{0, l_{sj}^q - h_{tj}^T, l_{tj}^T - h_{sj}^q\}. \tag{7}$$

Empty blocks have zero counts and zero intervals, and we explicitly set $G_{st} = 0$ whenever either block is empty. For positions x and y in nonempty blocks s and t, respectively, the reverse triangle inequality gives

$$\begin{aligned}\|x - y\|_2 &\geq \left|\|x - p^{\mathrm{pr}}_{W,j}\|_2 - \|y - p^{\mathrm{pr}}_{W,j}\|_2\right| \\ &\geq \max\{0, l^q_{sj} - h^T_{tj}, l^T_{tj} - h^q_{sj}\}.\end{aligned} \tag{8}$$

Taking the maximum over probes shows that G_{st} lower-bounds every point-pair distance in those blocks. For a nonnegative block-cost matrix A, write $\mathrm{DTW}(A)$ for the minimum summed cost along a path from its upper-left to lower-right cell, with the same three allowed steps. The coarse bound is $\mathrm{DTW}(G)$.

B. Weighted Bounds

For each query block, its smallest cost over candidate blocks lower-bounds the matching cost of each query position in that block. Weighting this minimum by the block's position count accounts for those positions. We subtract this minimum before processing candidate blocks so that their contributions do not count the same part of a cost twice:

$$\begin{aligned}u_s &= \min_t G_{st}, \qquad v_t = \min_s (G_{st} - u_s), \\ R_{st} &= G_{st} - u_s - v_t \geq 0, \\ L_{\mathrm{row}} &= \sum_s n_s u_s + \sum_t m_t v_t + \mathrm{DTW}(R).\end{aligned} \tag{9}$$

Here u_s is the query-block contribution per position, v_t is the candidate-block contribution after subtraction, and R contains the remaining costs. Its DTW term retains the order of blocks. The subscript `row` means that query blocks (matrix rows) are processed first.

Processing candidate blocks (matrix columns) first gives a second bound:

$$\begin{aligned}v'_t &= \min_s G_{st}, \qquad u'_s = \min_t (G_{st} - v'_t), \\ R'_{st} &= G_{st} - u'_s - v'_t \geq 0, \\ L_{\mathrm{col}} &= \sum_s n_s u'_s + \sum_t m_t v'_t + \mathrm{DTW}(R').\end{aligned} \tag{10}$$

We keep the strongest of the coarse and two weighted bounds:

$$L_{\mathrm{block}} = \max\{\mathrm{DTW}(G), L_{\mathrm{row}}, L_{\mathrm{col}}\}. \tag{11}$$

C. Validity

Proposition 3: In real arithmetic, $0 \leq L_{\mathrm{block}}(q, T) \leq D_W(q, T)$.

Proof: Fix any original alignment π. For a matched index pair $(i, r) \in \pi$, let $s(i)$ and $t(r)$ be the blocks containing positions p^q_i and p^T_r. The block-cost inequality gives

$$\begin{aligned}\sum_{(i,r)\in\pi} \|p^q_i - p^T_r\|_2 &\geq \sum_{(i,r)\in\pi} G_{s(i),t(r)} \\ &= \sum_{(i,r)\in\pi} \big(u_{s(i)} + v_{t(r)} + R_{s(i),t(r)}\big).\end{aligned} \tag{12}$$

Every query position is matched at least once. Since $u_s \geq 0$, the sum of query contributions is at least $\sum_s n_s u_s$. Similarly, $v_t \geq 0$ gives a candidate contribution of at least $\sum_t m_t v_t$. These sums count different portions of each matched-pair cost, not two copies of that cost.

Map the alignment to its block pairs and collapse consecutive repetitions of the same pair. Consecutive blocks preserve the original order, so this produces a valid block path. Removing repetitions cannot increase the sum of the nonnegative remaining costs R_{st}. Its cost is at least $\mathrm{DTW}(R)$, the minimum over all such paths. Consequently,

$$\begin{aligned}\sum_{(i,r)\in\pi} \|p^q_i - p^T_r\|_2 &\geq \sum_s n_s u_s + \sum_t m_t v_t \\ &\quad + \mathrm{DTW}(R) = L_{\mathrm{row}}.\end{aligned} \tag{13}$$

The inequality holds for every original alignment, hence $L_{\mathrm{row}} \leq D_W(q, T)$. Reversing the subtraction order proves $L_{\mathrm{col}} \leq D_W(q, T)$. Applying the same block-path argument directly to G proves $\mathrm{DTW}(G) \leq D_W(q, T)$. Taking their maximum preserves the lower-bound inequality.

For trajectories with fewer than four positions, the mapped path ends at the last nonempty block pair. Extend it to the final padded cell through zero-cost cells. On every padded cell, $G_{st} = 0$ and the nonnegative decomposition forces $u_s = v_t = R_{st} = 0$; the column-first case is identical. Thus padding adds no cost, and the argument still applies. ∎

Filtering can therefore stop early when $L_{\mathrm{block}}(q, T) > \widehat{r}_{W,q}$. Equality does not trigger rejection. Remaining candidates proceed to original DTW refinement with its running-bound early stopping.

a) Implementation: The block-filter implementation provides `counts` and `stats_torch` for the block partition, `bound_torch` for the reference calculation, and `pair_kernel` for the GPU calculation. The existing implementation also applies a dtype-dependent downward numerical guard; the proof above concerns the mathematical bound in real arithmetic.

Appendix D
DTW Signal Lower Bound

This is the signal bound used to rank DTW candidates (`eq:dtw-signal`), not the block bound in Appendix C or the learned summary retrieval score.

A. Definition

Let $A = (x_1, \ldots, x_n)$ and $B = (y_1, \ldots, y_m)$ be nonempty trajectories, and let $c_{W,1}, \ldots, c_{W,M_s}$ be a shared set of $M_s \geq 1$ probes. For a vertex x, its signal at probe j is $\sigma_j(x) = \|x - c_{W,j}\|_2$. Define the signal matching cost by

$$\ell_{\mathrm{sig}}(x, y) = \max_{1\leq j\leq M_s} |\sigma_j(x) - \sigma_j(y)|. \tag{14}$$

Let $\Pi_{n,m}$ contain the monotone paths from $(1, 1)$ to (n, m) with steps $(1, 0)$, $(0, 1)$, or $(1, 1)$. Both quantities use this same path set, retaining every vertex in its original order:

$$\begin{aligned}L^{\mathrm{sig}}_W(A, B) &= \min_{\pi\in\Pi_{n,m}} \sum_{(i,l)\in\pi} \ell_{\mathrm{sig}}(x_i, y_l), \\ D_W(A, B) &= \min_{\pi\in\Pi_{n,m}} \sum_{(i,l)\in\pi} \|x_i - y_l\|_2.\end{aligned} \tag{15}$$

Here DTW sums Euclidean distances, without squaring or length normalization. Earlier manuscript notation D_W^{sig} denotes the same signal quantity as L_W^{sig} above.

B. Validity

Proposition 4: For any shared probe positions, in real arithmetic, $0 \le L_W^{\mathrm{sig}}(A, B) \le D_W(A, B)$.

Proof: The Euclidean triangle inequality gives, for every probe j,

$$\begin{aligned} \sigma_j(x) &\le \|x - y\|_2 + \sigma_j(y), \\ \sigma_j(y) &\le \|x - y\|_2 + \sigma_j(x). \end{aligned} \tag{16}$$

Combining these two inequalities yields $|\sigma_j(x) - \sigma_j(y)| \le \|x - y\|_2$. The right-hand side does not depend on j, so taking the maximum over probes preserves it: $0 \le \ell_{\mathrm{sig}}(x, y) \le \|x - y\|_2$.

Now choose an original-DTW minimizing path $\pi_W \in \Pi_{n,m}$, which exists because the trajectories and path set are finite. The signal minimum cannot exceed the signal cost along this particular path. Summing the point-pair inequalities along it gives

$$\begin{aligned} 0 \le L_W^{\mathrm{sig}}(A, B) &\le \sum_{(i,l) \in \pi_W} \ell_{\mathrm{sig}}(x_i, y_l) \\ &\le \sum_{(i,l) \in \pi_W} \|x_i - y_l\|_2 \\ &= D_W(A, B). \end{aligned} \tag{17}$$

The signal-minimizing path need not equal π_W. No interchange of the minimum, sum, or maximum is used. Repeated vertices, ties, and single-vertex trajectories do not change the argument. Learning can change the bound's tightness, but validity holds for any shared probe positions. ∎

a) Implementation: The signal encoder `encode` computes vertex-to-probe distances. The `signal_kernel` takes the maximum absolute signal difference for each matched pair and applies the three-predecessor DTW recurrence. The `pairs` function buckets trajectory lengths without dropping vertices or changing the recurrence. The proof concerns exact real arithmetic, not a guarantee that floating-point roundoff preserves the inequality without a numerical guard. It does not make the separate learned summary score a lower bound.

Repository

The README proofs and corresponding implementation are available at https://github.com/DongXinfzu/LEAP.